

NANYANG TECHNOLOGICAL UNIVERSITY

SINGAPORE

Table 4 Source Library

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Lin Qianhui

Soong Qi Rong

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Part A: Source Library

Source Library link:
https://vertexaisearch.cloud.google.com/global/home/cid/39bc6644-69da-426c-8971-bbfc89f33c6a/r/notebook/c809989b-d7ac-4221-8161-94980b3e322d

Source	Author/Year/Date	Disciplinary lens	Role in Knowledge Base	Contributor
1	Tan, Y. W., Ngu, R. W. Y., et al. (2025). *Profiles and well-being of older adult caregivers in Singapore*. Singapore Management University. https://ink.library.smu.edu.sg/rosa_reports/29/	Social Sciences & Humanities	This source shows that more older adults are becoming caregivers in Singapore and face distinct health and caregiving challenges, while also highlighting gaps in their access to available support resources.	Kang Xun
2	Vaingankar, J. A., Asharani, P. V., Tan, B., Abdin, E., Wang, P., Jeyagurunathan, A., Chua, B. Y., Yao, F., Mahendran, R., Ng, L. L., Chong, S. A., & Subramaniam, M. (2026). Caregivers' burden and psychiatric morbidity among informal caregivers of older adults: Results from a national survey. *BMC Geriatrics, 26*(1), Article 1015. https://doi.org/10.1186/s12877-026-07739-6	Social Sciences & Humanities	This source shows that Singaporean caregivers, comprising mostly older caregivers, have higher rates of psychiatric morbidity. The effect is more prominent in caregivers that helped with basic self-care tasks.	Marsya
3	Tan, C., Chee, W. X., Chin, S. F., Ong, H., Duhalde, M., & Teoh, S. (2026, May 20). *Seniors*	Social Sciences & Humanities	This source shows how typically seen challenges faced by caregivers are more	Kayla

	caring for seniors: What caregiving looks like in a super-aged society. https://www.straitstimes.com/multimedia/graphics/2026/05/senior-caregivers/index.html		pronounced for older caregivers, as well as how it has impacted their own health.	
4	Loo, Y., Yan, S., & Low, L. (2021). Caregiver burden and its prevalence, measurement scales, predictive factors and impact: a review with an Asian perspective. *Singapore Medical Journal*, *63*(10). https://doi.org/10.11622/smedj.2021033	Social Sciences & Humanities	This source shows that intensive caregiving can worsen caregivers' physical and psychological wellbeing, highlighting how demanding caregiving responsibilities may compound existing health challenges older caregivers already face.	Zach
5	Yong, T., Visaria, A., & Malhotra, R. (2025). Impact of Informal Caregiving at Older Ages on Loneliness and Social Networks in Singapore. *Gerontology*, *71*(11), 990–1002. https://doi.org/10.1159/000548267	Social Sciences & Humanities	This source shows that older caregivers in Singapore experience greater loneliness, highlighting an overlooked psychosocial burden and the need for accessible support that addresses caregivers' social and emotional wellbeing.	Kayla
6	Lim-Soh, J., Sung, P., Quach, H.-L., & Malhotra, R. (2024). Sharing in Caring: Family Caregiving Task-Sharing Patterns for Older Adults in Singapore. *The Journals of Gerontology, Series B: Psychological Sciences and*	Social Sciences & Humanities	This source investigates family task-sharing patterns for elderly care, where older caregivers frequently shoulder primary duties alone, highlighting how relying strictly on informal family structures fails to prevent	Kang Xun

	Social Sciences, *80*(1). https://doi.org/10.1093/geronb/gbae186		severe strain.	
7	Zhou, Y., Bai, Z., Wan, K., Qin, T., He, R., & Xie, C. (2024). Technology-based interventions on burden of older adults' informal caregivers: a systematic review and meta-analysis of randomized controlled trials. *BMC Geriatrics*, *24*(1). https://doi.org/10.1186/s12877-024-05018-w	STEM	This source identifies ways technology-based interventions alleviate caregiver burdens, demonstrating how elderly caregivers can benefit via reducing caregiving tasks, providing information and training, facilitating care coordination, and offering psychosocial support.	Qi Rong
8	Liu, Q., Xia, L., Wu, C., Dang, W., Li, J., Liu, Y., Shang, C., Deng, Y., & Zhang, J. (2026). Caregiving experiences and support challenges of older caregivers from the perspective of the digital divide: a qualitative study. *Frontiers in Public Health*, *14*. https://doi.org/10.3389/fpubh.2026.1797984	STEM	This source shows that older caregivers' concerns about technology's reliability, safety and effectiveness can create distrust, potentially hindering their willingness to adopt digital health technologies for caregiving.	Qi Rong
9	Visaria, A., Aithal, S., & Malhotra, R. (2023). Digital technology use, in general and for health purposes, by older adults in Singapore. *Aging and Health Research*, *3*(1), 100117. https://doi.org/10.1016/j.ahr.2023	STEM	This source identifies affordability and complexity of technological solutions, together with age-related functional decline (vision impairment) being key factors influencing older caregivers'	Qi Rong

	.100117		usage of technology.	
10	Gao, J., et al. (2025). Developing a multicomponent intervention for family caregivers of older adults in Singapore: A Delphi study. *Health & Social Care in the Community*. https://doi.org/10.1155/hsc/8495394	STEM	This source identifies digital resources containing localized caregiving information as useful to address needs, also acknowledging that teaching older caregivers these technologies might be the biggest challenge.	Zach
11	Hassan, A. Y. I. (2020). Challenges and Recommendations for the Deployment of Information and Communication Technology Solutions for Informal Caregivers: Scoping Review. In JMIR aging (Vol. 3, Issue 2, p. e20310). PubMed. https://doi.org/10.2196/20310	STEM	This source highlights how older caregivers with limited digital capabilities may be inefficient when using e-health solutions to relieve burdens, when it requires learning technical skills or using complex interfaces.	Kang Xun
12	Smith, M. L., Lee, S., Neelamegam, M., Vollmer Dahlke, D., Southerland, J. L., Baker, Z. G., Ma, K. P. K., Petrovsky, D. V., Rahemi, Z., Sefcik, J. S., Bacsu, J.-D. R., Kew, C. L., & Ory, M. G. (2025). Perceived technology usefulness for caregiving among unpaid caregivers: a National Cross-Sectional Study. In	STEM	This source shows that older caregivers perceive caregiving technology as less useful, suggesting they may not recognise its potential benefits for reducing caregiving burden, which limits technology adoption.	Kayla

	Frontiers in public health (Vol. 13, p. 1578701). PubMed Central. https://doi.org/10.3389/fpubh.2025.1578701			
13	Seah, S. J., Mahirah, D., Ho Zhong Hao, C., Tan Yan Fang, C., Si Yinn, L., Yi, X., Koh Wan Cheen, C., Yong Ling, N., Chao Min, T., & Lian Leng, L. (2025). Unmet Needs Among Older Adult Informal Caregivers and Care Recipients in Singapore: A Qualitative Study. Journal of Aging & Social Policy, 1–19. https://doi.org/10.1080/08959420.2025.2475265	Governance	This source shows information gaps that persist among older caregivers regarding financial and caregiving support provided through government channels. Governments need accessible approaches to connect older caregivers with available support.	Qianhui
14	*ACTION PLAN FOR SUCCESSFUL AGEING 2023*. (2023). Ministry of Health. https://www.moh.gov.sg/others/resources-and-statistics/action-plan-for-successful-ageing/	Governance	The government runs the Caregiver Support Action Plan and Seniors Go Digital separately, the latter only equipping seniors with general-purpose digital skills with no equivalent offering for caregiving-specific technology needs	Marsya

15	Zhan, S., & Huang, L. (2023). State familism in action: aging policy and intergenerational support in Singapore. China Population and Development Studies, 7, 111–129. https://doi.org/10.1007/s42379-023-00132-5	Governance	This source shows the Singapore government shaping caregiving to be a state-family collaboration, and how policies must work in tandem with elderly caregivers for them to fulfil their caregiving responsibilities.	Marsya
16	Moriyama, T. (2026). Informal caregiving and labor supply at older ages: Heterogeneity by firm-size before age 60. The Journal of the Economics of Ageing, 34, 100618. https://doi.org/10.1016/j.jeoa.2026.100618	Business	This source shows frequent and intensive informal caregiving responsibilities are associated with reduced employment among older caregivers, highlighting how caregiving demands can constrain their ability to remain in the workforce.	Zach
17	Azman, N. D. B., Visaria, A., Goh, V. S., Østbye, T., Matchar, D., & Malhotra, R. (2024). Informal caregiving time and its monetary value in the context of older adults in Singapore. *Aging and Health Research, 4*(2), Article 100193. https://doi.org/10.1016/j.ahr.2024.100193	Business	This source shows that the monetary value of informal caregiving time for Singaporeans is significant for older caregivers, showing how caregiving can take time away from work, exacting economic costs.	Qianhui
18	Keita Fakeye, M. B., Samuel, L. J., Drabo, E. F., Bandeen-Roche, K., & Wolff, J. L. (2023). Caregiving-Related Work	Business	This source shows how caregiving can cause employees, including older caregivers, to miss work or	Qianhui

	Productivity Loss Among Employed Family and Other Unpaid Caregivers of Older Adults. *Value in Health*, *26*(5), 712–720. https://doi.org/10.1016/j.jval.2022.06.014		perform less productively, reducing organisational productivity and creating economic costs for businesses.	

Part B: Problem Statement

Gone are the days where we were an 'aging society', Singapore today has transitioned into a "super-aged" nation, with 1 in 4 Singaporeans projected to be 65 and above by 2030. Amidst this silver tsunami, more caregivers are older adults. Notably, the median caregiver age for those providing care daily is 62 (Tan et al., 2025).

Current literature shows older caregivers facing unique challenges. Many older caregivers treat caregiving as a personal responsibility, and do not delegate or seek help for it (Tan et al., 2026; Lim-Soh et al., 2024). Taking on most caregiving tasks consequently leads to their feelings of loneliness and stress (Yong et al., 2025), which takes a toll on their mental health (Vaingankar et al., 2026). Their finances are strained due to retirement, from the need to juggle multiple responsibilities (Moriyama, 2026). Physically, caregiving strains their bodies already weakened by age. Already affected by chronic health conditions (Tan et al., 2025), older caregivers struggle with tasks like mobility assistance, which increases their risks of injury (Loo et al., 2021).

Fortunately, Singapore treats caregiving as a state-family collaboration (Zhan & Huang, 2023), with numerous support schemes available (Ministry of Health, 2023). Nonetheless, due to poor technological literacy, many older caregivers remain unaware of these avenues available online (Seah et al., 2025). This also manifests in misunderstandings and distrust towards technology's utility and reliability (Smith et al., 2025), and inability to overcome steep learning curves of using complex caregiving technology that could potentially ease their caregiving load (Zhou et al., 2024; Hassan, 2020). Functional declines prove to be significant barriers to technological adoption too (Visaria et al., 2023). The gap thus lies in making caregiving technology more user-friendly and accessible for older caregivers to harness its benefits.

(248 words)

Part C: Critical AI Reflection

a) Insight

One instance (Figure 1) where Gemini Notebook helped us uncover an unexpected insight was when we were trying to consolidate the different issues involving older caregivers. Initially, the sources spanned different domains, and covered trends that seemed isolated. For instance, trends highlighted were older caregivers facing mental and physical stress, their technology uptake being low and the government providing caregiving support. To have a clearer understanding of the bigger picture, we uploaded our sources into Gemini, and explicitly instructed it to cross-analyse sources. Gemini's output pointed out what we had missed - no trend was independent of others, and relationships could be established between issues. In particular, Gemini connected the dots in the following way: government support information were mostly on websites which were complicated to access, and due to unfamiliarity with technology, it made these supports practically inaccessible to older caregivers, who experienced more physical and mental stress from caregiving responsibilities. Such rigorous analysis would have been difficult alone, requiring us to identify trends from each source individually, and form numerous relationships not explicitly written out, between them. Gemini greatly cut time spent on identifying patterns, leaving more time to find hidden nuances in sources.

b) Failure

A failure occurred earlier in our research, when we prompted Gemini Notebook to summarise different burdens faced by older caregivers. Until then, our source library contained a variety of studies discussing challenges faced by caregivers in general, instead of specific ones faced by our older target group. Rather than flagging a lack of evidence, Gemini compiled the issues discussed across the sources, without considering their relevance to older caregivers. For example, it identified loss of productivity and impacts on career development as key financial problems, although these concerns were more relevant to younger caregivers just entering the workforce (Figure 2). Older caregivers instead are likely approaching or have reached retirement, highlighting Gemini's flaw in generalising findings. We learnt that sources provided to Notebook must be relevant to our target group, so that sufficient context

is provided to Gemini to generate a logically coherent response, and better prompting must be conducted to minimise errors.

c) Intervention

To correct this, we refined the source library by removing articles irrelevant to older caregivers, before re-prompting Gemini to compile challenges specific to older caregivers, and identify instances of insufficient context. Consequently, Gemini scoped the challenges, one being age-related functional decline. Gemini also highlighted assumptions that required further consideration, such as usage of the MediSave scheme to reduce older caregivers' financial strain. Gemini pointed out that its effectiveness could vary depending on financial circumstances, noting MediSave contributions to be affected by income (Figure 3). It made us realise identifying a problem specific to our target group does not mean it affects members to similar extents. We delved deeper into issues with this insight, and strove to uncover nuances that would be lost by treating older caregivers as a single homogenous group.

d) Contradiction

While analysing our sources, we identified a contradiction between Zhou et al. (2024) and Liu et al. (2026). Zhou's systematic review found that technology-based interventions could reduce caregiver burden, while Liu's qualitative study identified issues like digital literacy could increase their burdens. We prompted Gemini to analyse the contradiction, while taking their different methodologies and outcome measures into account. Gemini disregarded neither source, and instead, reconciled the findings by distinguishing between effectiveness and barriers. A crucial limitation in our initial framing emerged: demonstrating that a technology can reduce burden does not necessarily mean that it will be effective for older caregivers in reality. We therefore became more attentive during our research to the conditions surrounding technology adoption, particularly whether barriers are present impending its intended users.

(600 words)

Appendix A: Chat log with Gemini

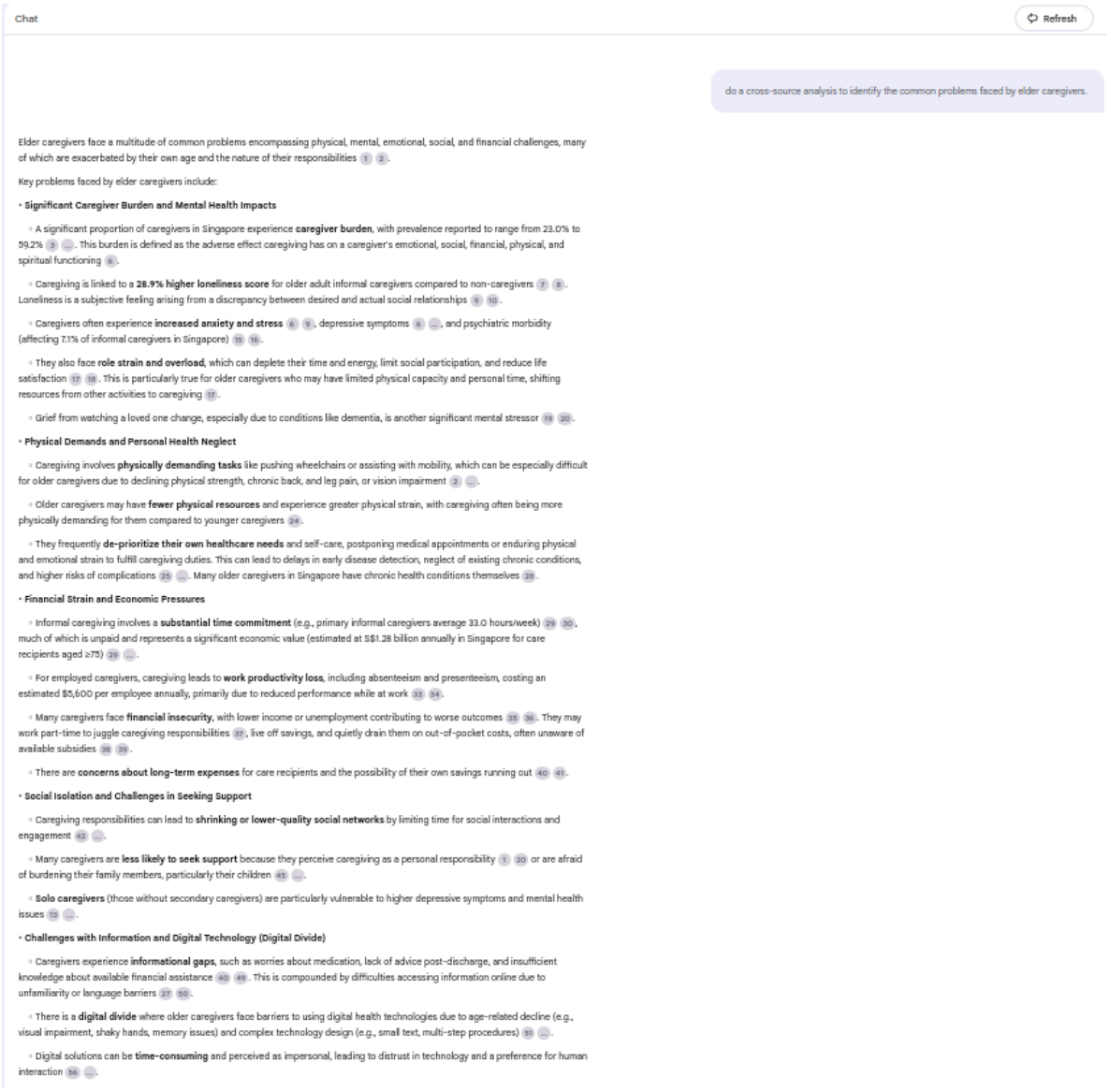


Figure 1: Gemini helping us develop stronger insights

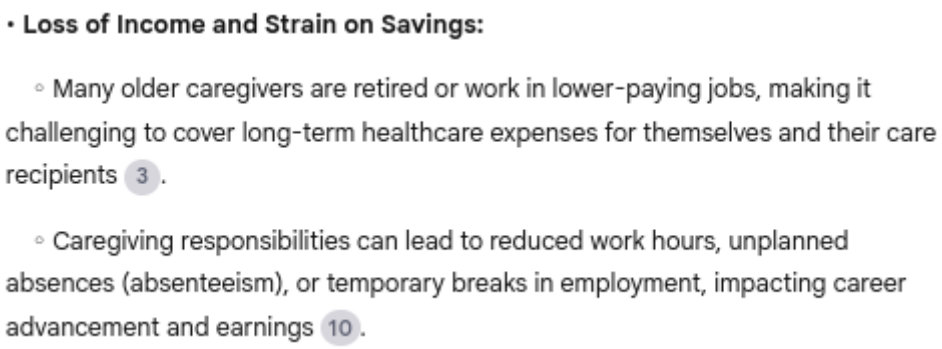


Figure 2: Gemini erroneously identifying career advancement as a significant problem facing older caregivers

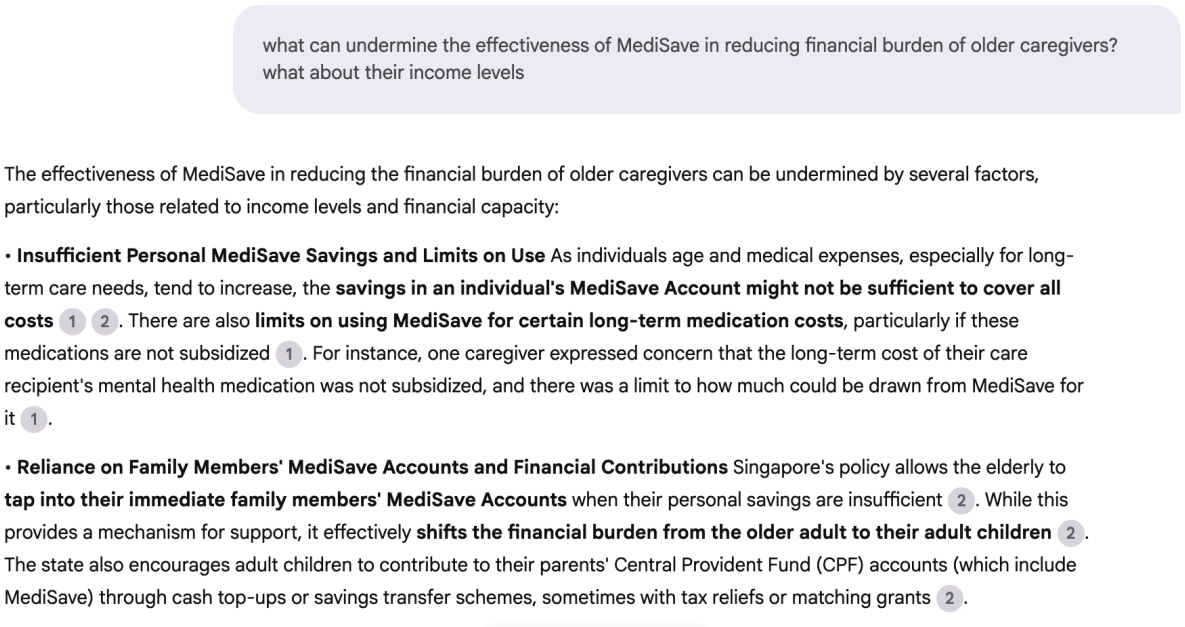


Figure 3: Gemini identifying assumptions that can undermine MediSave's effectiveness